Jyotirmoy chakraborty

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
MSR Energy Sciences	Indian Institute of Technology, Guwahati	8.7 (Current)	2025–Present
B.Tech Mechanichal	Tezpur University	7.33	2019–2023
Senior Secondary	CBSE Board	85.8%	2017
Secondary	ICSE Board	86.33%	2015

Projects

Safety-Gated Event-Triggered Edge–Cloud Digital Twin for Lithium-Ion Battery Management (MTP) *2026–Present*
Python, MATLAB/Simulink, EKF, PINN, PyTorch, PyBaMM, OPAL-RT, TI C2000, Battery Modelling *GitHub*

- Built an edge–cloud Battery Management System (BMS) digital twin for SOC, SOH, and RUL estimation using physics-based battery models, AI models, and event-triggered communication.
- Reduced cloud communication by approximately **82.8%** compared with an always-update EKF while maintaining low SOC estimation error in simulation.
- Prepared a MATLAB/Simulink and OPAL-RT-based hardware-in-the-loop (HIL) workflow for real-time battery plant validation and embedded deployment.
- Exported model assets for embedded targets, including ONNX models and C header weight files for TI C2000 and STM32 implementation.

BatteryWise – Local Battery Analytics Desktop Application *2026*
Python, FastAPI, SQLite, PyWebView, Battery Telemetry, Data Analytics *GitHub*

- Built a local-first desktop application for monitoring laptop battery telemetry, including charging state, cycle count, voltage, temperature, battery percentage, and estimated capacity.
- Developed a telemetry logging pipeline using FastAPI and SQLite to support local battery behaviour analysis, event tracking, and exportable battery-health datasets.
- Implemented raw and daily CSV export modules to support future machine-learning-based battery degradation forecasting and battery-health trend analysis.

NEXUS P2P *2024*
Electron, Node.js, Capacitor, Vanilla JavaScript, HTML/CSS *GitHub*

- Engineered a decentralized, local-first workspace application enabling zero-configuration peer discovery through custom UDP broadcasting and active subnet scanning.
- Implemented end-to-end encryption for messaging and clipboard synchronization using Elliptic Curve Cryptography (`secp256k1`) and AES-256-GCM.
- Developed a real-time reactive user interface using HTTP Server-Sent Events (SSE) and native Node.js streams for instant state synchronization and memory-efficient transfer of large files and directories.

Real-Time Social Distance Monitoring (BTP) *Jun 2022–Mar 2023*
YOLOv5, OpenCV, Python, Jetson Nano *GitHub*

- Developed a real-time person-detection system using YOLOv5, achieving **89% accuracy** in crowded indoor settings.
- Applied perspective transformation and camera calibration to estimate distances between individuals.
- Designed a custom 3D-printed camera mount, reducing hardware cost by **60%**.

Technical Skills

- **Energy Systems & Battery Modelling:** Battery Management Systems (BMS), Lithium-Ion Batteries, SOC Estimation, Electro-Thermal Modelling, Equivalent Circuit Models, Digital Twin, Energy Storage Systems, Distributed Energy Resources
- **AI/ML & Data Analytics:** PyTorch, TensorFlow, Scikit-learn, Regression, Time-Series Analysis, Anomaly Detection, Model Evaluation, CNNs/ResNet, LLM Integration, Prompt Engineering
- **Hardware & Engineering Tools:** Arduino, Sensor Data Acquisition, Fusion 360, Basic Embedded Systems, Battery Telemetry Logging,Opal-RT , Imperix , TI C2000

Key Courses Taken

- **Energy Sciences:** Renewable Energy Systems, Energy Storage Systems, Distributed Energy Resources, Fundamentals of Energy Engineering, Operation and Instrumentation,Carbon Accounting: GHG Emissions Calculation Scope 1 & 2 , Scope 3 Carbon/GHG emissions

Positions of Responsibility

- **Student Coordinator**,Center for Career Development, IIT Guwahati *Sept. 2025 – May 2026*

Achievements

- **GATE 2023**,Secured 96.66 percentile, top 3% in the country *2023*